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ITC-Suite_GDAL

Description of the GDAL version of the ITC-Suite

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Introduction

The ITC-Suite was first developed for the ARIES image analysis system in the 90's.

Later, it mostly ran under PCI/XPACE, then PCI/EASI, with PCI/ImageWorks and PCI/Geomatica
to display and deal with PCI's ".pix" files (PCI v2022 at this time)

Instead of a PCI file containning all, it is now assumed that all will be in a single "project directory": images as tif files, bitmap as 1-bit tif files, vector layers as shp files, signatures as discrete files, etc., and that ArcGIS (or other software) may be used to view all these.

In order to deal with less items on the command line, file nomenclature is often predefined, but often using a "base file name" from the main image file.

Having an image file name such as "BaseName_IRGB.tif" for example will ensure that other files will be of the type "BaseName_aaaa.ext"
For examples, "BaseName_VFOL.tif" as output from itcvfol_g.exe
or "BaseName_ISOL.tif" as output from itcisol_g.exe

Of course the "BaseName" should reflect the name of the main project in that directory.

The underscore is used to separate the "base file name" from the rest of the file name.

Full directory path is now mostly supported (i.e., not fully tested yet as the philosophy

was to do everything in the same directory from a "cmd windows"). However, it was needed

for the integration of the GDAL version of the Suite used with a ArcGIS Toolbox (GUI).

The GDAL version of the ITC-Suite can be used from ArcGIS or ArcGISPro via appropriate

toolboxes, allowing users to stay completely within that environment.

NOTE: Partial use of a PCI file is often possible (as input). However, user entered segment numbers will be that of GDAL, not of PCI.

Use "gdalinfo" and "ogrinfo" on the PCI file to find that order.

(e.g., bitmaps are numbered after images, vector layers are separate, signature segments are not visible, ...)

NOTE: The "ITC-Suite Manual" and "ITC-Suite.chm" are still very relevant to all this

but should be used as guides, not gospel.

NOTE: My GDAL programs always complain when overwriting an existing output file.

Most of the time, it is not an issue. However, sometimes when ArcGIS has control

of the file, it may not overwrite at all and leave you with previous results

Acknowledgment to GDAL (Geospatial Data Abstraction Library)

- GDAL - Geospatial Data Abstraction Library: Version 2.1.1 (July 2016)
- GDAL - Geospatial Data Abstraction Library: Version 3.0.0 (Dec. 2019)
- GDAL - Geospatial Data Abstraction Library: Version 3.3.3 (Dec. 2021)
- GDAL - Geospatial Data Abstraction Library: Version 3.11 (Feb. 2025)

Open Source Geospatial Foundation,

Thanks Frank (Warmerdam)

Running GDAL version of the ITC-Suite

From command prompt window

To run the ITC-Suite from a Windows 10/11 "cmd prompt window", see **ITC-Suite_GDALv3.10_Run.txt**.

Copy and paste the three(3) SET lines into your window. **Note:** They may need modifications if the GDAL or ITC-Suite locations are different on your computer.

From ArcGIS Pro

To run ITC-Suite programs from the ArcGIS Pro environment (Here for GDAL version 3.10.0) :

One needs to load the appropriate ToolBox (e.g., ITC-Suite_GDAL-v3-10.atbx) AND first run "GDAL_V3_10_Setup" before anything else.

This will mimic the above three lines for the ArcGIS environment.

For LinuxLinux

To run the ITC-Suite from a Linux "bash shell" check: [Linux_GDAL_ITC-Suite_Compile_v3.txt](#) It also has the setup for compiling programs.

Note: This is very exploratory at this point in time. However, programs run under Linux/Mint that I have used via Oracle Virtual Box

Additional Info

For info on how to run the ITC-Suite, one should check the "ITC-Suite_Docs" directory above.

For additional info on how to run the ITC-Suite, one should check the ITC-Suite Manual from 2010 (meant for the PCI environment).

It is highly recommended to read the **first 12 pages** to get a good overview of the ITC-Suite.

Gougeon, F.A. 2010. [The ITC Suite Manual : A Semi-Automatic Individual Tree Crown \(ITC\) Approach to Forest Inventories](#). Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C. Canada. June 2010. 92 p.